

Homework 01

Question 1 (Least squares and matrix calculations)

Set the random seed to 43201. Generate $n = 100$ observations with $p = 5$ predictors according to

$$x_{ij} \stackrel{\text{iid}}{\sim} N(0, 1), \quad \varepsilon_i \stackrel{\text{iid}}{\sim} N(0, 0.7^2),$$

with all predictors and errors generated independently. Let

$$\mathbf{X} = [\mathbf{1} \quad \mathbf{x}_1 \quad \cdots \quad \mathbf{x}_5], \quad \beta = (3, 1.4, -0.9, 0.7, 0, 1.1)^\top,$$

and generate

$$\mathbf{y} = \mathbf{X}\beta + \varepsilon.$$

- a. Generate the data and report the dimensions and rank of $\mathbf{X}$. Calculate the least-squares estimate $\hat{\beta}$ using a numerically stable least-squares routine, without explicitly calculating $(\mathbf{X}^\top \mathbf{X})^{-1}$. Compare the estimates with the coefficients used to generate the data. “{r} set.seed(43201) n = 100 p = 5

```
x = matrix(rnorm(n * p), nrow = n, ncol = p) colnames(x) <- paste0("x", seq_len(p))
```

```
X = cbind("(Intercept)" = 1, x) beta <- c( "(Intercept)" = 3, x1 = 1.4, x2 = -0.9, x3 = 0.7, x4 = 0, x5 = 1.1 )
```

```
epsilon = rnorm(n, mean = 0, sd = 0.7) y = as.vector(X %*% beta + epsilon)
```

```
dim(X) qr(X)$rank
```

```
fit = lm.fit(x = X, y = y) beta_hat = fit$coefficients fitted_values = as.vector(X %*% beta_hat) residuals = y - fitted_values
```

```
coefficient_comparison = data.frame( coefficient = names(beta), truth = unname(beta), estimate = unname(beta_hat) )
```

b. Calculate the fitted values, the residual vector $\mathbf{r} = \mathbf{y} - \mathbf{X}\widehat{\boldsymbol{\beta}}$, and the training root mean squared error

```

$$
\operatorname{RMSE}
=
\left(
\frac{1}{n}\sum_{i=1}^nr_i^2
\right)^{1/2}.
$$

```{r}
training_rmse = sqrt(mean(residuals^2))
normal_equation_error = max(abs(crossprod(X, residuals)))

print(dim(X))
print(qr(X)$rank)
print(coefficient_comparison)
print(training_rmse)
print(normal_equation_error)

```

c. Report  $\|\mathbf{X}^T \mathbf{r}\|_\infty$ . Explain why this value should be close to zero and what it tells us about the residual vector.

The training RMSE is about 0.6764 so this means that the model's predictions differ from the observed values by about 0.6764 units on average. The residuals are orthogonal to every column of the matrix and the maximum absolute value is about  $5.9 \times 10^{-14}$ .

## Question 2 (Gaussian likelihood for linear regression)

Continue with the data from Question 1. Suppose

$$\mathbf{Y} \mid \mathbf{X} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I}_n).$$

The log-likelihood, including the constant term, is

$$\ell(\boldsymbol{\beta}, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}).$$

a. For fixed  $\sigma^2$ , explain why maximizing  $\ell(\boldsymbol{\beta}, \sigma^2)$  over  $\boldsymbol{\beta}$  is equivalent to minimizing the residual sum of squares. What does this imply about the maximum-likelihood estimate of  $\boldsymbol{\beta}$ ?

The only part of the log-likelihood that changes with  $\boldsymbol{\beta}$  is the residual sum of squares. Because it is subtracted, decreasing the residual sum of squares increases the log-likelihood. By maximizing the likelihood gives the same estimate

of  $\beta$  as minimizing the residual sum of squares. Therefore, it is the least-squares estimate from Question 1.

- b. For fixed  $\beta$ , differentiate the log-likelihood with respect to  $\sigma^2$  and derive its maximum-likelihood estimate. Calculate this estimate at  $\hat{\beta}$  and compare it with the unbiased estimate of  $\sigma^2$ . Explain why the two denominators differ.

```
“{r} rss = sum(residuals^2) sigma2_mle = rss / n sigma2_unbiased = rss / (n
- ncol(X))
```

```
log_likelihood = function(beta_value, sigma2, X, y) { residual = y - as.vector(X
%% beta_value) - nrow(X) / 2 log(2 * pi * sigma2) - sum(residual^2) / (2 *
sigma2) }
```

They are different because MLE divides the residual sum of squares by  $n=100$  while the unbiased estimate divides by 94 ( $n-6 = 94$ ). Six comes from the one intercept and 5 predictor coefficients. The unbiased estimator adjusts for the six degrees of freedom by using a smaller denominator.

c. Let

```
$$
\mathbf{e}_{x_2} = (0, 0, 1, 0, 0, 0)^T.
$$
```

Using the variance estimate from part b, plot

```
$$
\ell(\widehat{\beta} + t \mathbf{e}_{x_2},
\widehat{\sigma}_{\text{MLE}}^2)
\text{right)}
$$
```

over a grid of  $t$  values from  $-1.25$  to  $1.25$ . State where the maximum occurs and explain the result.

```
“{r}
e_x2 <- c(0, 0, 1, 0, 0, 0)
t_grid <- seq(-1.25, 1.25, length.out = 251)
profile_log_likelihood <- vapply(
 t_grid,
 function(t) {
 log_likelihood(
 beta_hat + t * e_x2,
 sigma2_mle,
```

```

 X,
 y
)
},
numeric(1)
)

print(c(
 sigma2_mle = sigma2_mle,
 sigma2_unbiased = sigma2_unbiased,
 maximizing_t = t_grid[which.max(profile_log_likelihood)]
))

plot(
 t_grid,
 profile_log_likelihood,
 type = "l",
 lwd = 2,
 col = "#1F5A94",
 xlab = expression(t),
 ylab = "Log-likelihood"
)
abline(v = 0, lty = 2, lwd = 2, col = "#C84A16")

```

The maximum occurs at  $t = 0$ . When  $t = 0$ , the coefficient equals the least-squares estimate. If  $t$  is anything other than 0, then the estimated coefficient  $x_2$  changes from the least-squares value. This therefore increases the residual sum of squares and decreases the log-likelihood.

### Question 3 (Starting values in nonconvex optimization)

Consider the function

$$f(x) = \exp(1.5x) - 3(x + 6)^2 - 0.05x^3,$$

with derivative

$$f'(x) = 1.5 \exp(1.5x) - 6(x + 6) - 0.15x^2.$$

- a. Plot  $f(x)$  over the interval  $[-40, 7]$ . Based on the plot, describe the important features of the objective function that may affect numerical optimization.

```
“{r} objective = function(x) { exp(1.5 * x) - 3 * (x + 6)^2 - 0.05 * x^3 }
```

```
gradient = function(x) { 1.5 * exp(1.5 * x) - 6 * (x + 6) - 0.15 * x^2 }
```

```
x_grid = seq(-40, 7, length.out = 1200)
plot(x_grid, objective(x_grid), type = "l", lwd = 2, col = "#13294B", ylim =
c(-450, 100), xlab = "x", ylab = "f(x)")
```

The function is nonconvex because of its two separate low points and local max in between. The function is also increasing very quickly due to the exponential term in the function. This affects the numerical optimization because the result is likely to depend on the optimizer's starting value.

b. Use BFGS to minimize  $f(x)$  twice, starting at  $x^{(0)} = -15$  and  $x^{(0)} = 0$ . Supply  $f'(x)$  to the optimizer. For each run, report the final value of

```
```{r}
starts <- c(-15, 0)
optimization_results <- lapply(
  starts,
  function(start) {
    optim(
      par = start,
      fn = objective,
      gr = gradient,
      method = "BFGS"
    )
  }
)

optimization_summary <- do.call(
  rbind,
  Map(
    function(start, result) {
      data.frame(
        starting_value = start,
        final_x = result$par,
        objective = result$value,
        absolute_gradient = abs(gradient(result$par)),
        converged = result$convergence == 0
      )
    },
    starts,
    optimization_results
  )
)

points(
```

```

optimization_summary$final_x,
optimization_summary$objective,
pch = 19,
col = c("#2F6FB3", "#C84A16")
)

print(optimization_summary)

```

When starting at $x = -15$, it converges to $x = -32.64911$ where the object value is -390.38577 . When starting at $x = 0$, it converges to $x = 2.34997$ where the object value is -175.86262 . Both absolute values of the derivative is close to 0 and optimizer reports a successful convergence. Both results are local minimums.

- c. Explain why the two runs can converge to different answers. Which run gives the lower objective value? What additional evidence would be needed before claiming that this point is the global minimum?

They run difference answers because the function has more than one local minimum, each starting value leads the optimizer towards a different low region. The run that gives the lower objective value is when it begins at $x = -15$. In order to support that this is the global minimum, one would need to double check that no lower points were missed. You could set all derivatives to 0 and compare each of the values.

Question 4 (Nearly collinear predictors)

Continue with $\mathbf{X}$ and $\mathbf{y}$ from Question 1. Set the random seed to 43202 and generate

$$u_i \stackrel{\text{iid}}{\sim} N(0, 1), \quad i = 1, \dots, n.$$

Define

$$\mathbf{x}_6 = \mathbf{x}_1 + 10^{-4}\mathbf{u}, \quad \mathbf{X}_+ = [\mathbf{X} \quad \mathbf{x}_6],$$

and let

$$\mathbf{y}^* = \mathbf{y} + 10^{-3}\mathbf{u}.$$

Use a numerically stable least-squares routine throughout.

```

“{r} set.seed(43202) u <- rnorm(n)
x6 <- x[, “x1”] + 1e-4 * u X_plus <- cbind(X, x6 = x6) y_star <- y + 1e-3 * u
fit_original <- lm.fit(x = X, y = y) fit_augmented <- lm.fit(x = X_plus, y =
y) fit_perturbed <- lm.fit(x = X_plus, y = y_star)

```

```

beta_augmented <- fit_augmentedcoefficientsbeta_perturbed <- -fit_perturbedcoefficients

fitted_augmented <- as.vector(X_plus %% beta_augmented) fitted_perturbed
<- as.vector(X_plus %% beta_perturbed)

condition_number <- function(design) { singular_values <- svd(design, nu = 0,
nv = 0)$d max(singular_values) / min(singular_values) }

fit_summary <- data.frame( design = c("X", "X_plus"), condition_number
= c( condition_number(X), condition_number(X_plus) ), training_rmse = c(
sqrt(mean(fit_originalresiduals^2)), sqrt(mean(fit_augmentedresiduals^2)) ) )

```

a. Fit $\mathbf{y}$ using $\mathbf{X}$ and $\mathbf{X}_+$. For each design matrix, report the

The original has a condition number of 1.5 and after adding x6, the condition number increases to about 20,000.

The training RMSE changes very little after adding x6, which means the model is not improving. The coefficients are 351.677 for x1 and -350.259 for x6.

The sum of these is 1.418 which is close to the original coefficient estimate for x1. This shows that separately the coefficients are unstable but together they are stable.

b. Fit $\mathbf{y}^*$ using $\mathbf{X}_+$. Report the changes in the coefficients of $\mathbf{y}^*$ value vectors using their root mean squared difference and maximum absolute difference.

After making the change, x1 decreases by about 10 while x6 increases by about 10. These are big changes to the coefficients with the small change that was made to them.

The fitted values barely changed. Their root mean squared difference is about 0.00102 and the max absolute difference is 0.00257. So, the large changes in individual coefficients do not really cause large changes in the model's predictions.

c. Use the identity

$$\begin{aligned}
 & \mathbf{y}^* - \mathbf{y} \\
 &= \\
 & 10(\mathbf{x}_6 - \mathbf{x}_1)
 \end{aligned}$$

to explain why the fitted values change very little while the two coefficients change substantially.

```

```{r}
coefficient_summary <- data.frame(
 response = c("y", "y_star"),
 beta_x1 = c(beta_augmented["x1"], beta_perturbed["x1"]),
 beta_x6 = c(beta_augmented["x6"], beta_perturbed["x6"])
)

coefficient_change <- (
 beta_perturbed[c("x1", "x6")] -
 beta_augmented[c("x1", "x6")]
)
fitted_difference <- fitted_perturbed - fitted_augmented

print(fit_summary)
print(coefficient_summary)
print(coefficient_change)
print(c(
 fitted_difference_rmse = sqrt(mean(fitted_difference^2)),
 fitted_difference_max = max(abs(fitted_difference))
))

```

Since x1 and x6 are almost the same, it is difficult for the model to determine how much of the effect belongs to each predictor. It is more reliable in predicting their combined effect than their individual effects. This means that the separate coefficient estimates should be interpreted cautiously.

## Question 5 (Data preparation and summary statistics)

The file `data/data-manipulation.csv` is a small constructed data table containing an observation identifier, a group label, two numeric predictors, and a response. Some response values are missing. For analyses involving the response, use only observations with a recorded response. Do not replace a missing response with zero or a group mean.

- a. Read the data and report its dimensions, column types, number of duplicated identifiers, and number of missing values in each column. Create an analysis table containing only observations with a recorded response, and define

```

```{r}
getwd() list.files(recursive = TRUE) data_file <- file.path("data",
"data-manipulation.csv") if (!file.exists(data_file)) { document_dir <-
Sys.getenv("QUARTO_DOCUMENT_PATH", unset = ".") data_file <-
file.path( document_dir, "data", "data-manipulation.csv" ) } small_data <-
read.csv("homework/week-01/data/data-manipulation.csv")

```

```
print(dim(small_data)) print(vapply(small_data, class, character(1))) print(c(
duplicated_identifiers = sum(duplicated(small_data$observation_id)) ))
print(colSums(is.na(small_data)))
```

```
$$
\texttt{feature\_sum}=\texttt{x1}+\texttt{x2}.
$$
```

There are 24 observations and five variables. The `observation_id` and `group` are stored as text while `x1`, `x2`, and `response` are numeric. There are no duplicated observation identifiers. The only variable with missing values is `response` with 3 missing entries. After removing those, there are 21 rows.

b. For each group, report the number of observations, the mean response, and the mean of `feature_sum`.

```
```\r}
analysis_data <- small_data[
 !is.na(small_data$response),
]
analysis_data$feature_sum <- (
 analysis_data$x1 + analysis_data$x2
)

group_summary <- do.call(
 rbind,
 lapply(
 split(analysis_data, analysis_data$group),
 function(group_data) {
 data.frame(
 group = group_data$group[1],
 observations = nrow(group_data),
 mean_response = mean(group_data$response),
 mean_feature_sum = mean(group_data$feature_sum),
 row.names = NULL
)
 }
)
)

largest_three <- analysis_data[
 order(analysis_data$response, decreasing = TRUE),
 c("observation_id", "group", "response", "feature_sum")
][1:3,]

number_recorded <- sum(!is.na(small_data$response))
```

```

stopifnot(
 nrow(analysis_data) == number_recorded,
 !anyNA(analysis_data$response),
 !anyDuplicated(analysis_data$observation_id)
)

print(group_summary)
print(largest_three)

```

Group	Observations	Mean response	Mean <code>feature_sum</code>
A	7	5.6000	3.6000
B	7	7.6029	4.9286
C	7	7.9971	6.5857

These summaries describe only observations with recorded responses. They should not be described as summaries of all 24 rows.

- c. Sort the analysis table by response from largest to smallest and report the first three rows, including `observation_id`, `group`, `response`, and `feature_sum`. Add code checks verifying that the number of retained rows equals the number of nonmissing responses in the original data, that the analysis table has no missing response values, and that its identifiers are unique.

The three largest recorded responses are:

<code>observation_id</code>	Group	Response	<code>feature_sum</code>
obs-22	C	9.24	6.6
obs-16	B	9.00	5.2
obs-14	B	8.66	5.0

## Question 6 (Can 39 million Fitbit records represent US adults?)

Patten et al. (2026) describe Fitbit data from 59,018 participants in the All of Us Research Program. The dataset spans 14 years and contains more than 39 million daily step records. Participants contributed data through one of two routes:

- Bring Your Own Device (BYOD): participants shared data from a Fitbit they already owned.
- Wearables Enhancing All of Us Research (WEAR): invited participants received a Fitbit at no cost.

In the general activity cohort, the BYOD and WEAR groups contained 32,035 and 22,474 participants, respectively. The BYOD value is listed first in each comparison below:

- 77.3% versus 55.1% reported being White;
- 6.1% versus 15.2% reported annual household income between \$10,000 and \$25,000; and
- median daily steps were 6,867 versus 5,797.

Suppose the target is the mean of the participant-specific average daily step counts among US adults during the study period, with each adult given equal weight. A researcher writes:

“This dataset contains more than 39 million daily Fitbit records. Therefore, the average of all recorded step counts should provide an accurate estimate of mean daily activity among US adults.”

- a. Are the 39 million daily records independent observations? What characteristics or behaviors could cause some participants to contribute more recorded days than others? Explain how averaging all recorded days would then weight participants unequally.

They are not independent because each participant contributes many days of data. Some people may contribute more days because they could have enrolled earlier or wore and synched their Fitbit more consistently. By averaging every recorded day gives these participants more weight than people with fewer recorded days.

- b. BYOD participants reported a median of 6,867 daily steps, compared with 5,797 among WEAR participants. Does this comparison show that already owning a Fitbit causes people to walk more? Explain your answer and give at least one plausible alternative explanation based on how participants entered the two groups.

No, the comparison does not prove that owning a Fitbit causes people to walk more. BYOD participants could already be more active, health-conscious, or interested in fitness. Since they were not randomly assigned, the difference could be caused by preexisting characteristics.

- c. A very large dataset can still produce a biased estimate of a population quantity when the people and observations entering the dataset are selected. Using this study, explain how selection of both participants and recorded days could make the observed data differ from the US adult population. Why might the naive average of all recorded daily step counts fail to estimate the stated target? You do not need to determine the direction of the bias.

The participants may not be a good representation of all US adults because Fitbit users may differ from the general population. Recorded days may also be selective because people do not wear or sync their devices everyday. In

addition to that, averaging all daily records gives more weight to participants with more data, so a large dataset can still produce a biased estimate.

**Source**

Patten, T., Preble, E. A., Master, H., et al. (2026). The All of Us Research Program's wearables dataset. *Nature Medicine*, 32, 2302-2310.